USED CAR PRICE PREDICTION

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Research question

The price of a used car depends on many factors, including its age, mileage, brand, engine type, power and condition. Buyers may find it difficult to determine whether a price is fair, while sellers need to set a price that is consistent with the market.

Our research question is: **Which vehicle characteristics best explain its price, and to what extent can the price be predicted from these characteristics?**

The project has two objectives:

- **Predictive objective:** build a regression model that estimates a vehicle's price with the lowest possible error.
- **Explanatory objective:** identify the variables that genuinely influence price and assess their relative importance.

Current progress

Completed: locating four datasets, loading them into a notebook, and carrying out an initial inspection (columns, first rows and descriptive statistics).
Still to do: variable selection, cleaning, construction of the final dataset, modelling and comparison of the two approaches. These tasks are presented below as our work plan; none has been carried out yet.

Data sources

No single dataset contains all the information we need. We therefore identified four real datasets on Kaggle from which we will build our own dataset. They come from different sources, and we will check for overlapping records during cleaning.

Dataset 1 — Used Car Price Prediction Dataset (used_cars.csv)

Source: https://www.kaggle.com/datasets/taeefnajib/used-car-price-prediction-dataset
Size: 4,009 rows, 12 columns. **Columns:** brand, model, model_year, milage, fuel_type, engine, transmission, ext_col, int_col, accident, clean_title, price.
Specific features: Cars.com listings, vehicles from 1974–2024, US market (prices in dollars and mileage in miles). Price and mileage are stored as text (e.g. "$10,300", "51,000 mi."). The engine field mixes power, displacement and engine configuration; clean_title contains missing values.

Dataset 2 — Car Price Prediction Dataset (car_price_dataset.csv)

Source: https://www.kaggle.com/datasets/sukhmandeepsinghbrar/car-price-prediction-dataset
Size: 10,000 rows, 10 columns. **Columns:** Brand, Model, Year, Engine_Size, Fuel_Type, Transmission, Mileage, Doors, Owner_Count, Price.
Specific features: Vehicles from 2000–2023; the only dataset containing the number of doors. Price and mileage units are unspecified. Values are unusually regular (mileage from 25 to around 300,000; prices from 2,000 to 18,301), and some combinations seem implausible, such as manual electric cars. We will assess whether this dataset is realistic or synthetically generated.

Dataset 3 — Honda Used Car Selling (honda_car_selling.csv)

Source: https://www.kaggle.com/code/themrityunjaypathak/car-price-prediction
Size: 999 rows, 6 columns. **Columns:** Year, kms Driven, Fuel Type, Suspension, Price, Car Model.
Specific features: Honda vehicles only, from 2000–2023, from the Indian market. Price values are text with units (e.g. "9.29 Lakh"), as are mileage values (e.g. "19006 kms"). The Suspension column actually describes the gearbox ("Automatic" or "Manual") and must be renamed.

Dataset 4 — 90,000+ Cars Data From 1970 To 2024 (CarsData.csv)

Source: https://www.kaggle.com/datasets/meruvulikith/90000-cars-data-from-1970-to-2024
Mirror: https://baselight.app/u/kaggle/dataset/meruvulikith_90000_cars_data_from_1970_to_2024
Size: 97,712 rows, 10 columns. **Columns:** model, year, price, transmission, mileage, fuelType, tax, mpg, engineSize, Manufacturer.
Specific features: Vehicles from 1970–2024, mostly around 2017. Currency and mileage units are unspecified, but tax and mpg suggest a UK market. Labels are inconsistent ("hyundi", lower-case "volkswagen", upper-case "BMW", "Semi-Auto"). Possible outliers include 470.8 mpg, zero engine displacement and zero tax.

WORK PLAN — NEXT STEPS

The four datasets differ in period, market, currency, units and available variables. They must therefore be harmonised before being combined.

1. Variable selection

For each dataset, we will retain the variables relevant to price: vehicle identity (brand and model), age and use (year and mileage), technical characteristics (fuel type, transmission and engine displacement), and—where available—history (number of previous owners and accident record). Columns with no plausible link to price, such as colours, will be excluded with justification.

2. Data cleaning

Cleaning has not yet been performed. We will apply it to each dataset before merging and document every decision:

- **Data types:** convert prices and mileage stored as text into numeric values (datasets 1 and 3).
- **Labels and units:** standardise column names (e.g. rename Suspension to transmission), categories (Petrol/Gasoline; Manual/Automatic/Semi-Auto), capitalisation and brand misspellings (dataset 4); convert miles and kilometres to a common unit and all prices to a common currency.
- **Duplicates:** remove identical rows, including records repeated across different datasets.
- **Outliers:** detect unusual values using the interquartile range (IQR) and consistency checks, such as 470.8 mpg or zero engine displacement in dataset 4.
- **Missing values:** remove rows if the amount of missing data is small; otherwise, impute using the median or mode, or use an “unknown” category (e.g. clean_title in dataset 1).

3. Building the final dataset

The merge has not yet been performed. We will combine the four cleaned datasets into one table with a consistent schema. Common matching fields will be brand, model, year and fuel type. In dataset 3, brand and model are combined in the Car Model field and will need to be separated. Variables unavailable in a dataset will be represented as missing values. We will also add a source-dataset column to check for source-related bias, especially because dataset 4 accounts for more than 80% of all rows.

4. Modelling

We will test two approaches:

- **Approach 1 — Feature selection:** train the model with several combinations of variables, then retain the best-performing and most relevant combination.
- **Approach 2 — Predefined features:** train models using a set of variables chosen in advance, based on our understanding of the domain.

5. Comparison

We will compare the results of the two approaches to determine whether exploring different feature combinations improves prediction compared with a reasoned, predefined selection, and whether both approaches identify the same key price determinants.

Data-quality consideration: Because the datasets come from different markets and may use different currencies or collection methods, we will carefully assess comparability before pooling them. The source indicator will help us identify possible dataset effects.